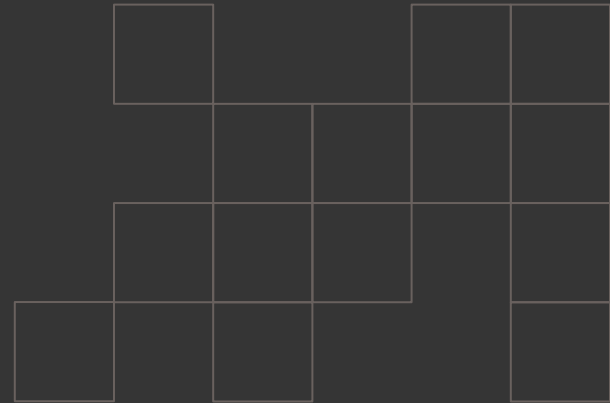
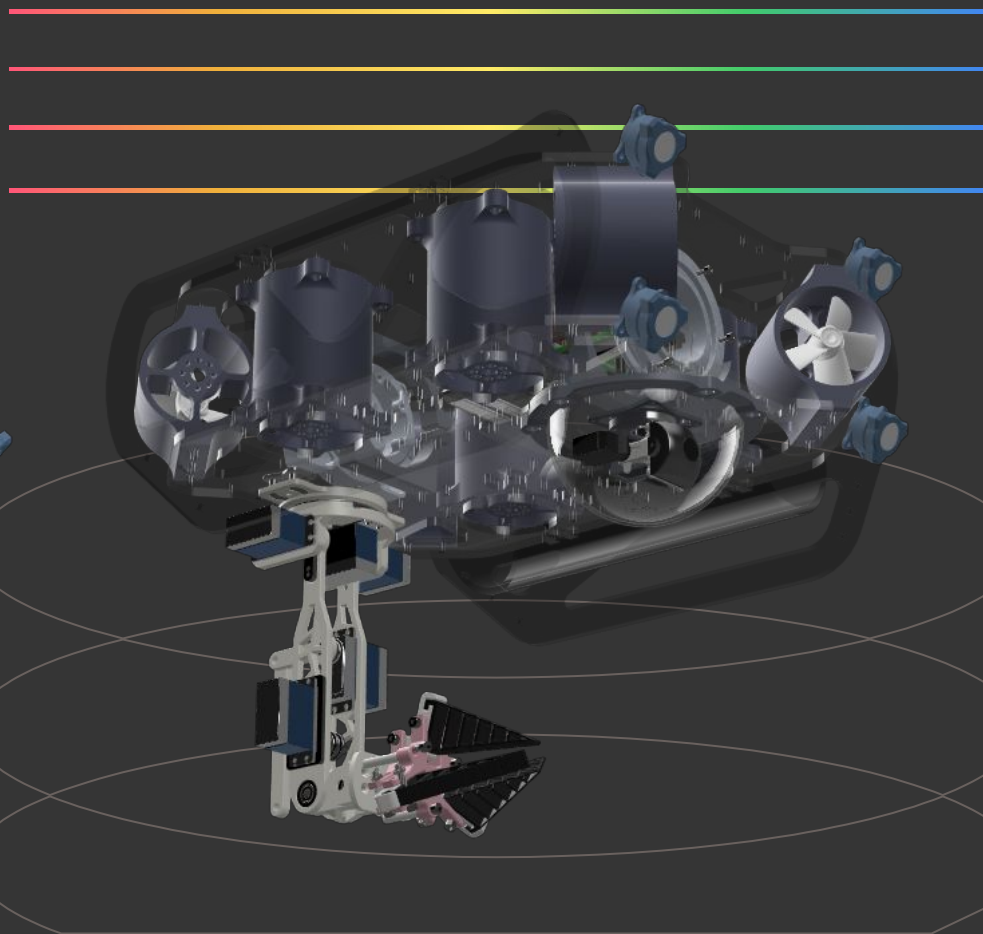
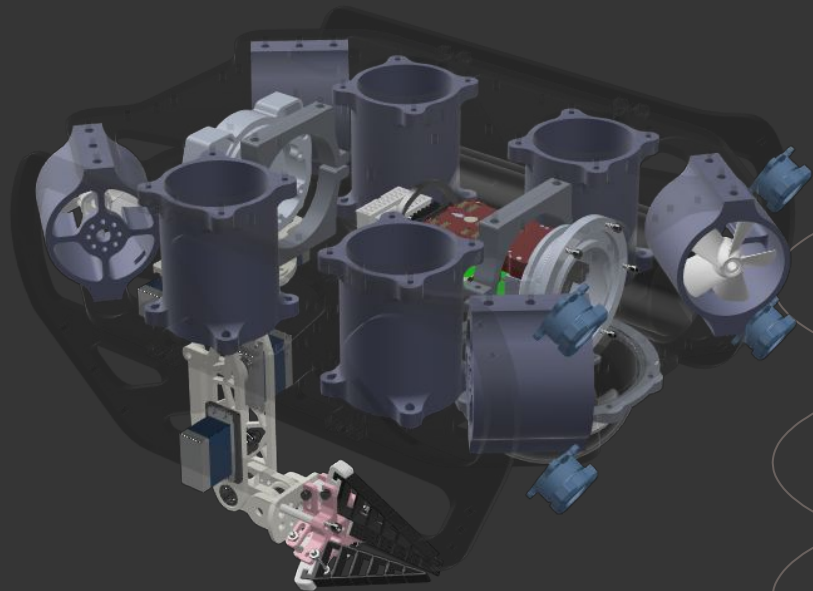


Submarine Project - Bluefin

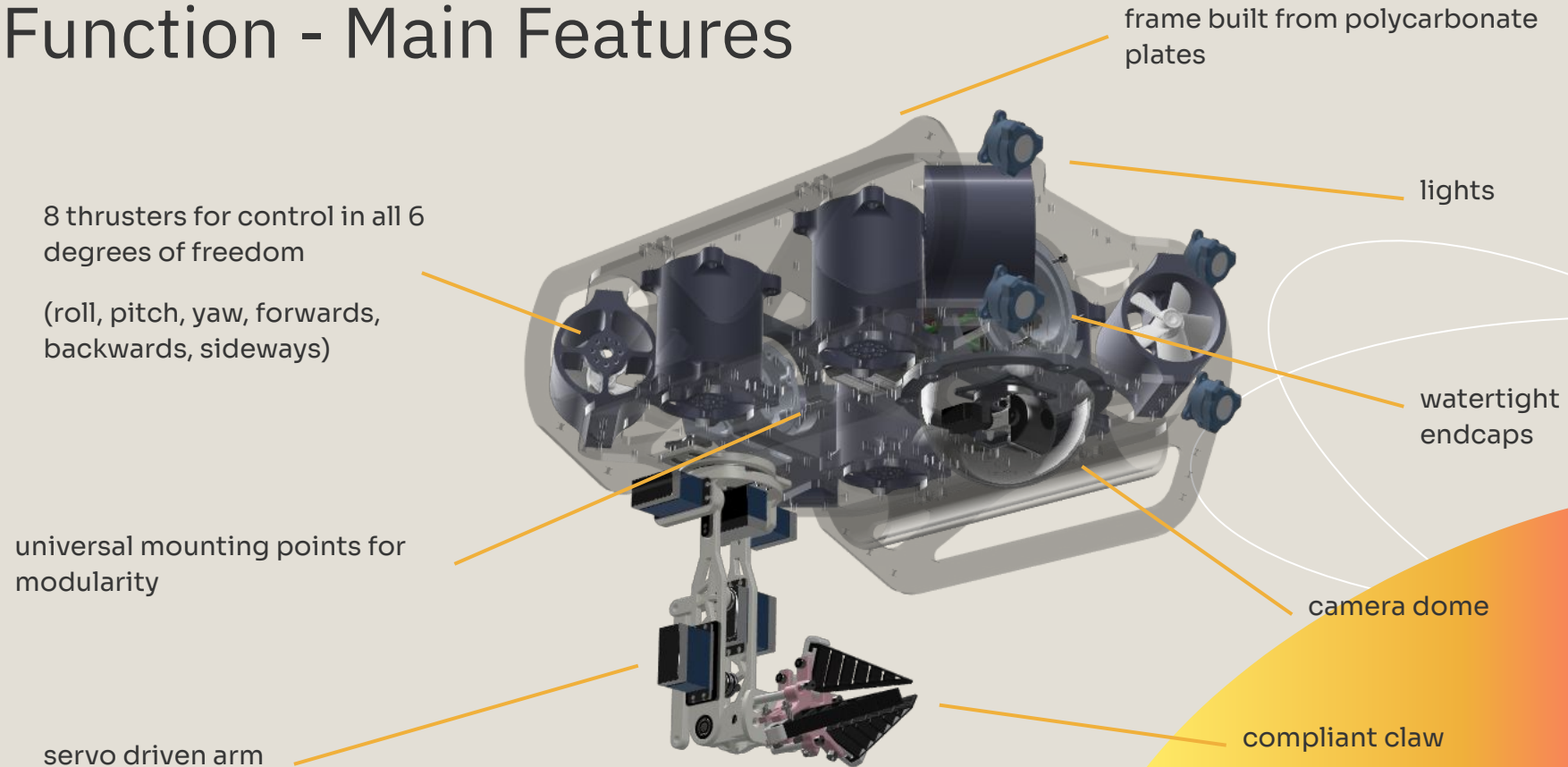


Bluefin

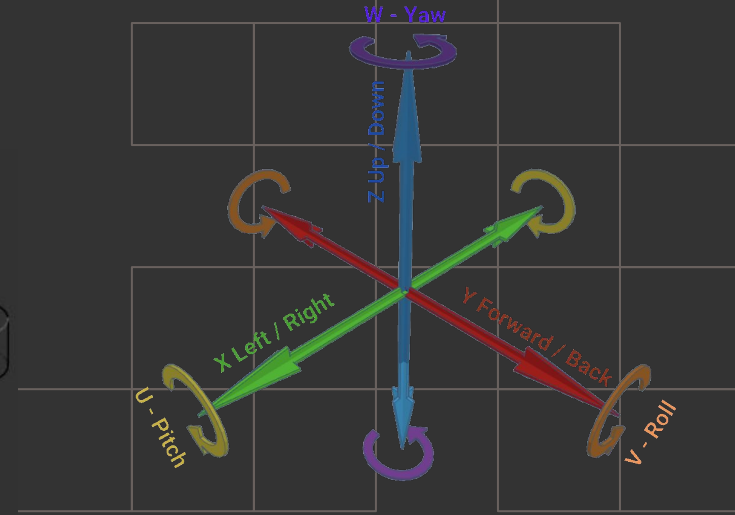
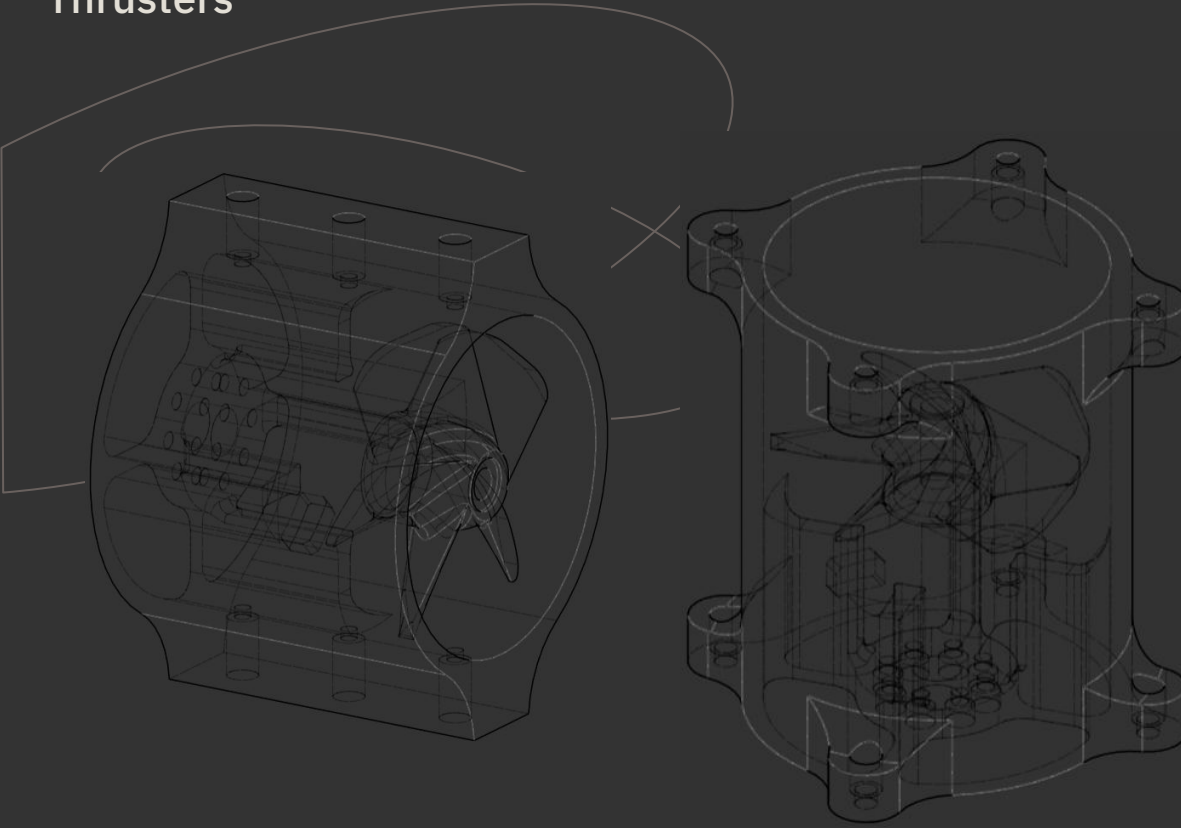
Submarine Overview



Function - Main Features



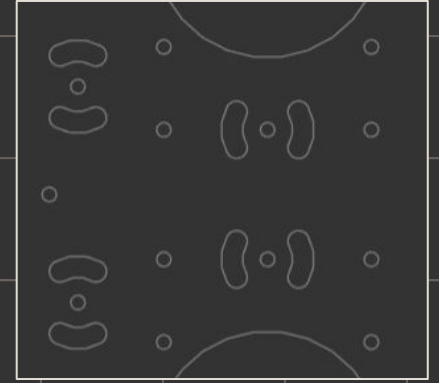
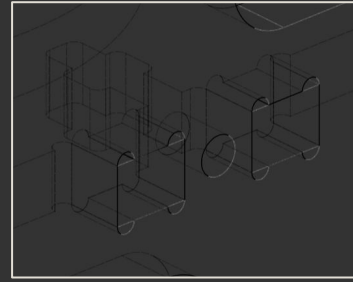
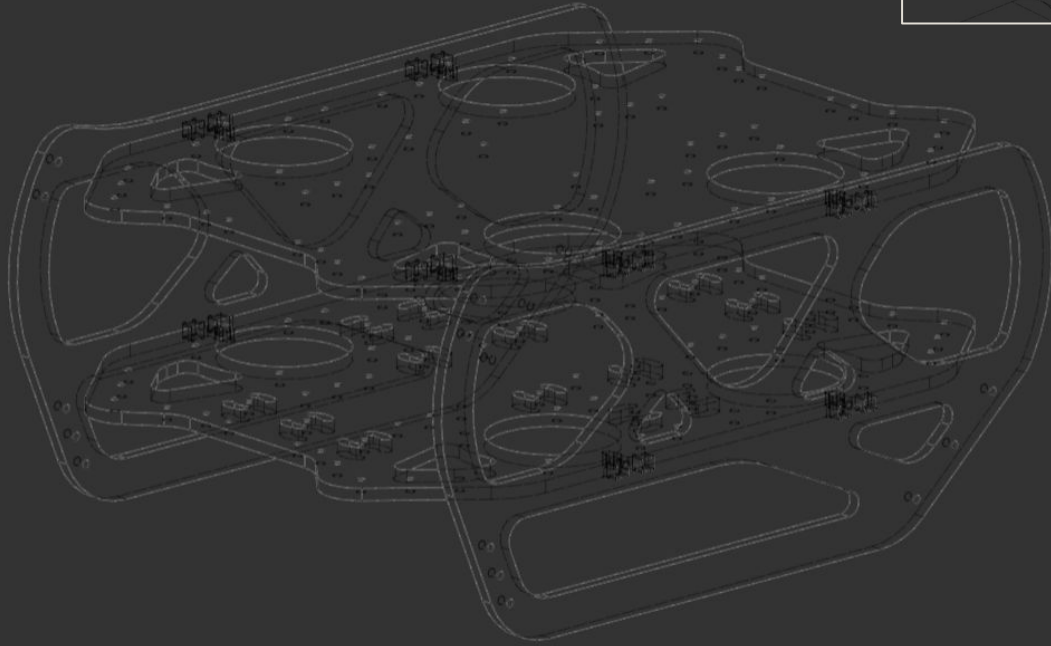
Thrusters



- each powered by a 2212 brushless motor
- 3d-printed
- mounted with headset inserts
- doubles as structure for the frame

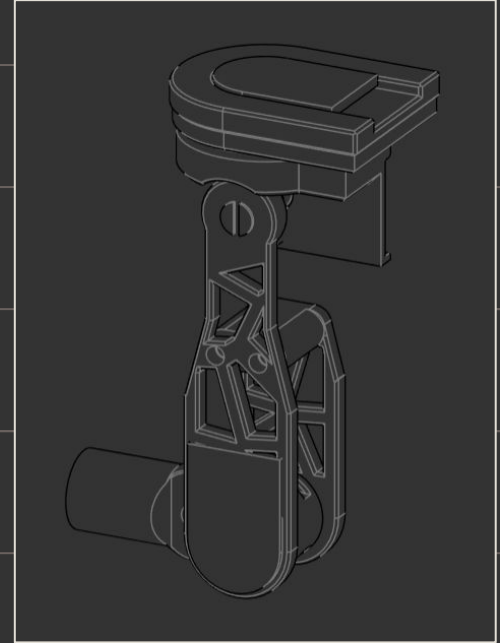
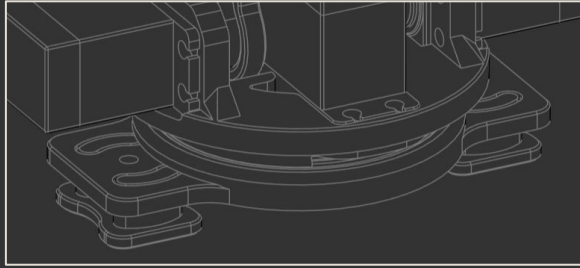
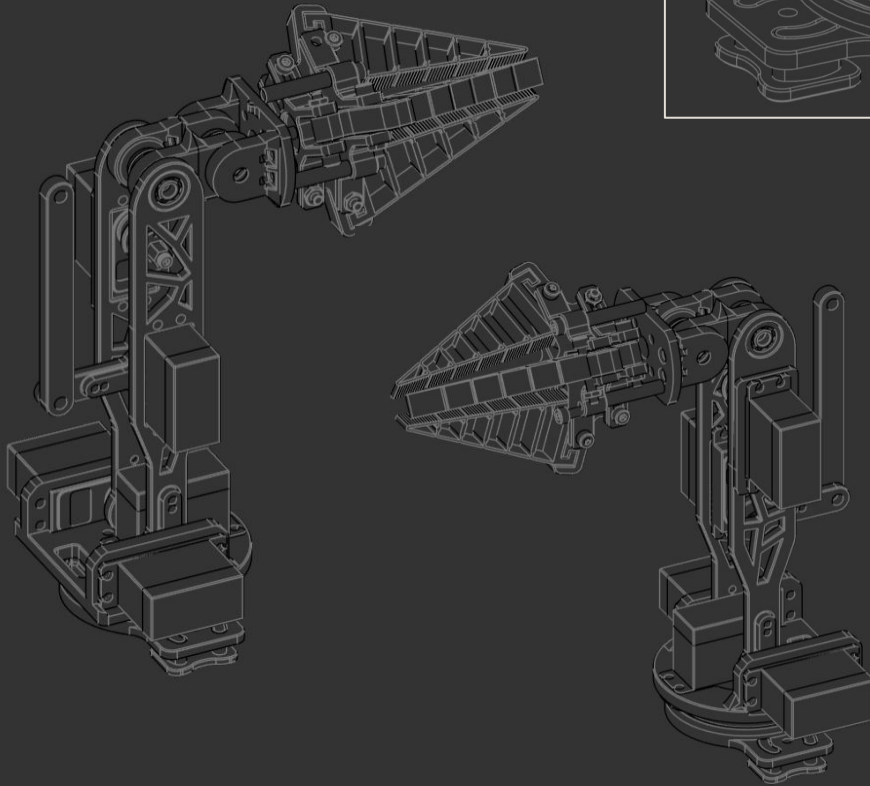
Bluefin

Polycarbonate Frame



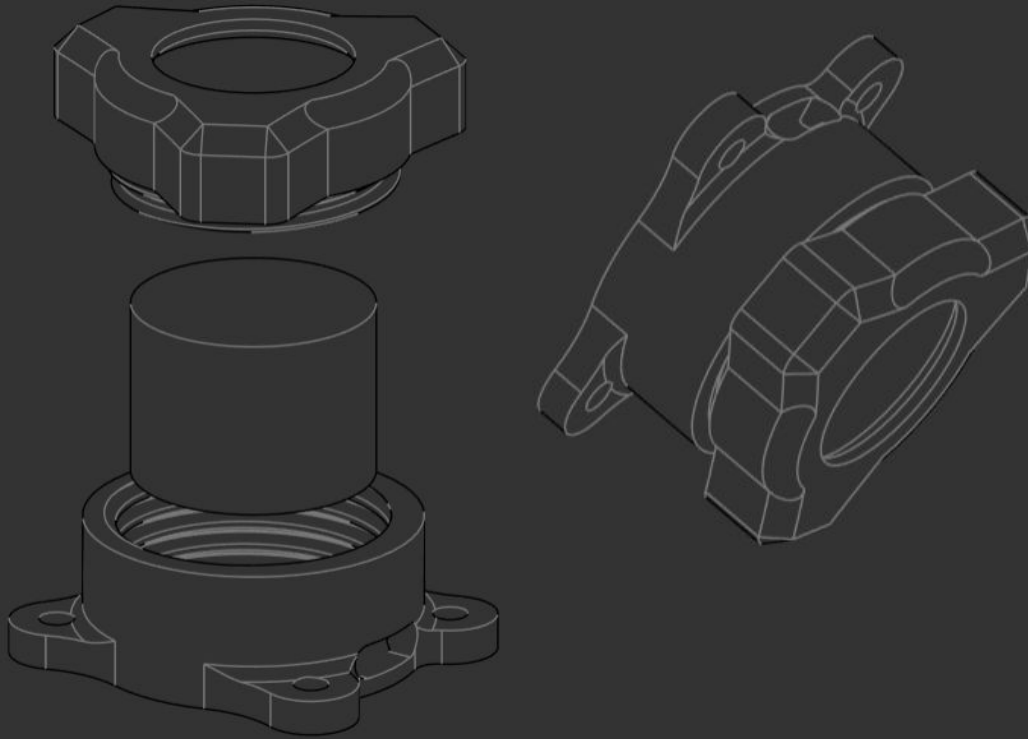
- built with 6mm polycarbonate plates
- serves as the mounting points for other subsystems
- universal mounting points for modules

Servo-Driven Arm

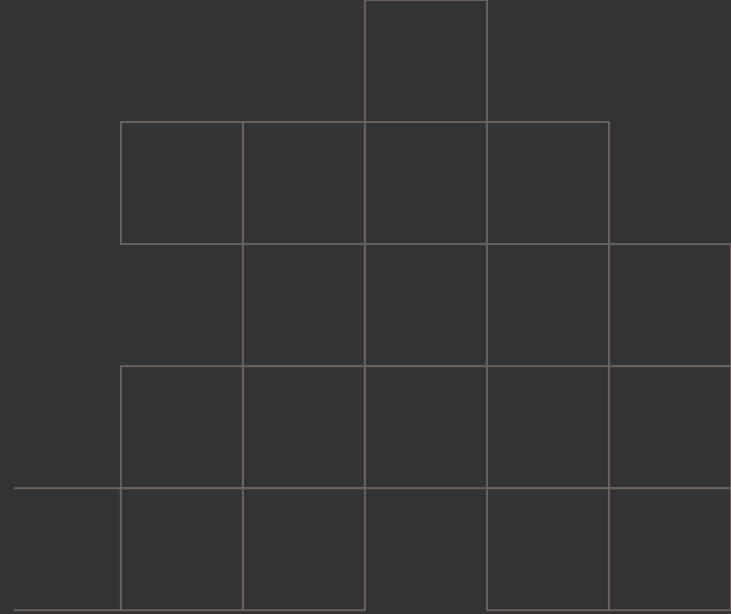


- 4 degrees of freedom driven with 5 waterproof servos
- mounted at the base with the universal mounting pattern
- controlled using a gesture control module

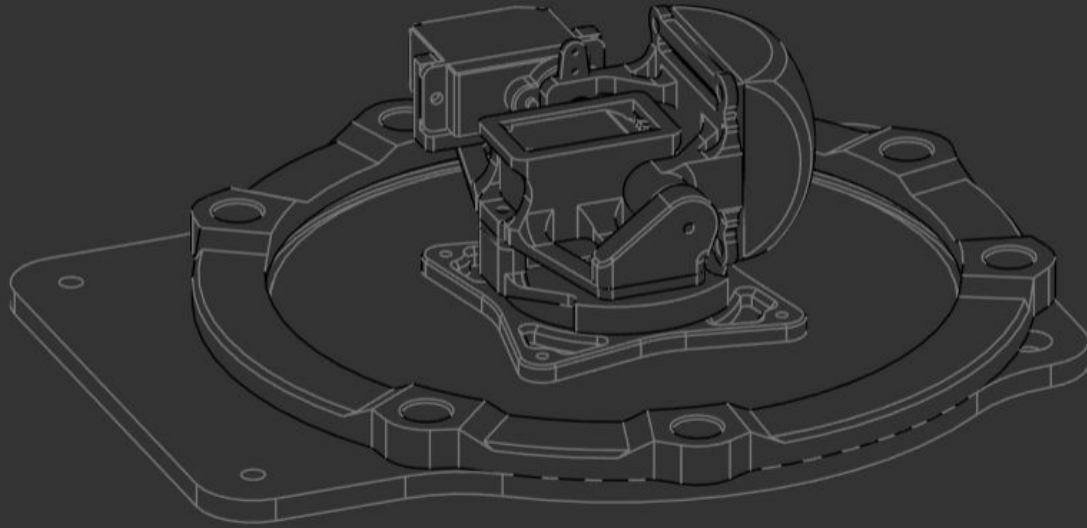
Lights



- simple 2 part construction
- top cover holds the light and diffuser in place
- both covers screw into each other using 3d printed threads
- sealed with epoxy

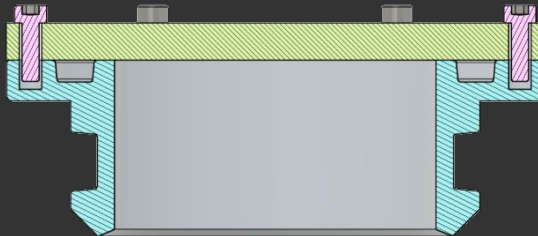
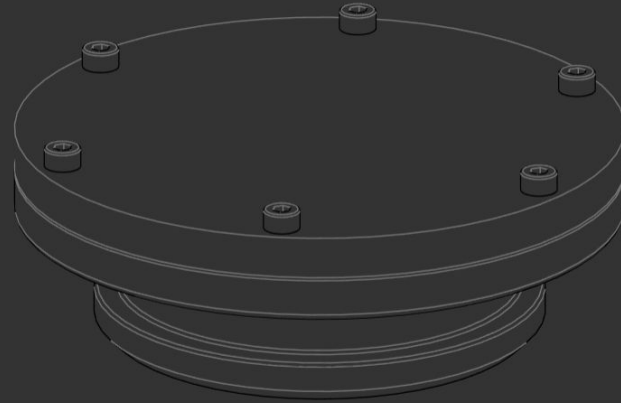
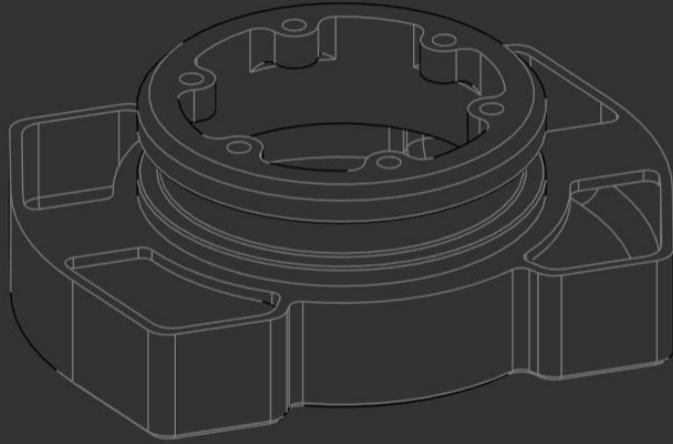


Camera Dome



- powered by two mini EMAX servos
- allows for 180 degrees of rotation in pitch axis and 360 degrees of rotation in yaw axis
- seal is made with an o-ring under the dome that's clamped down by the outer ring

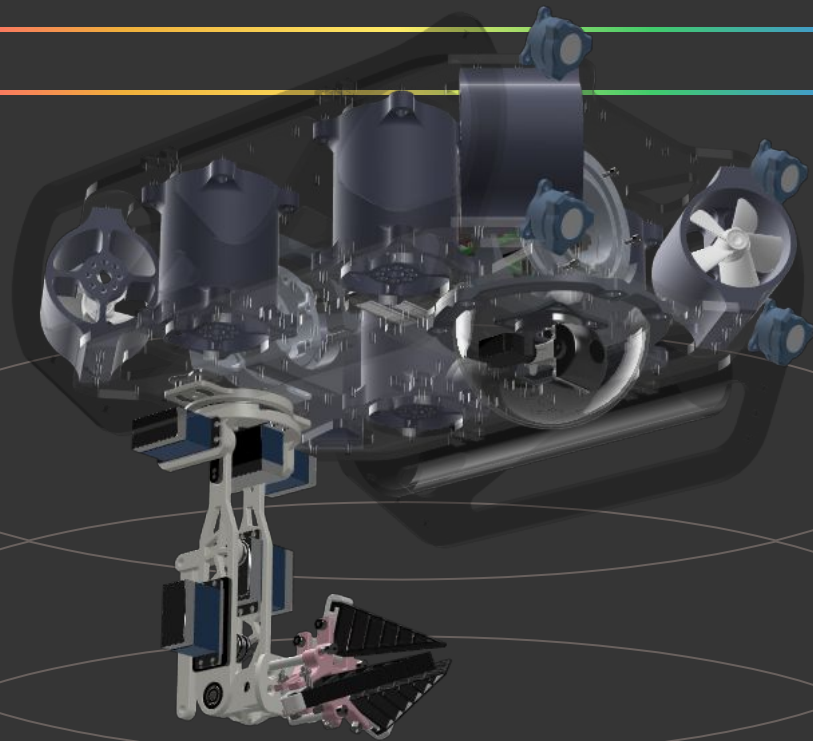
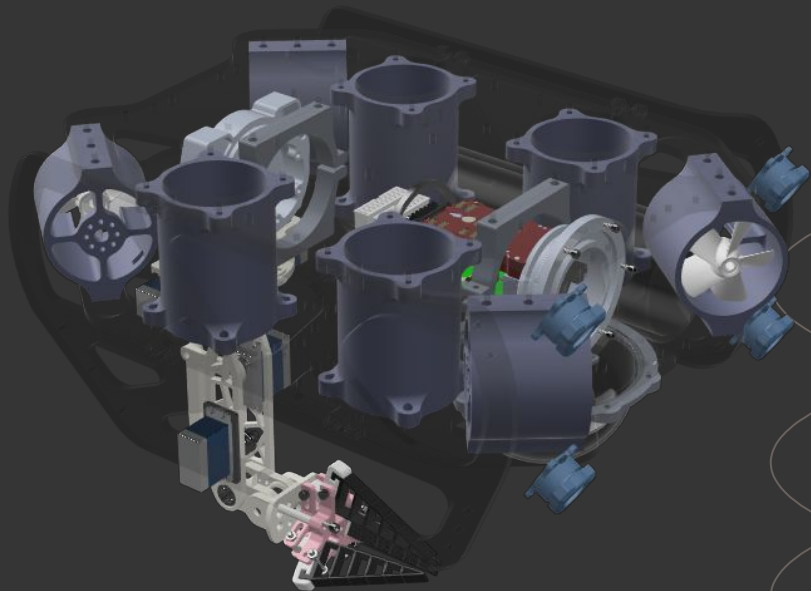
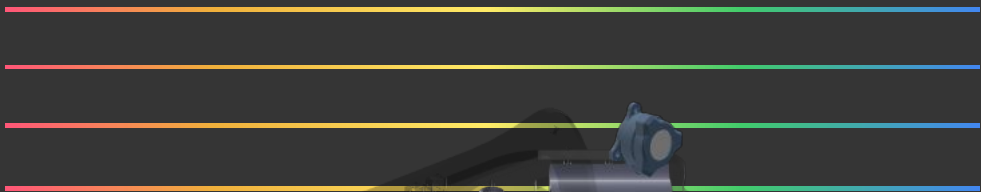
End Caps



- two endcaps for sealing the electronics tube from the front and back
- one side has an acrylic cover for the tether to pass through
- other side has cutouts for cables to pass through which will be sealed with epoxy
- 100% infill with optimized layer lines to prevent leaks

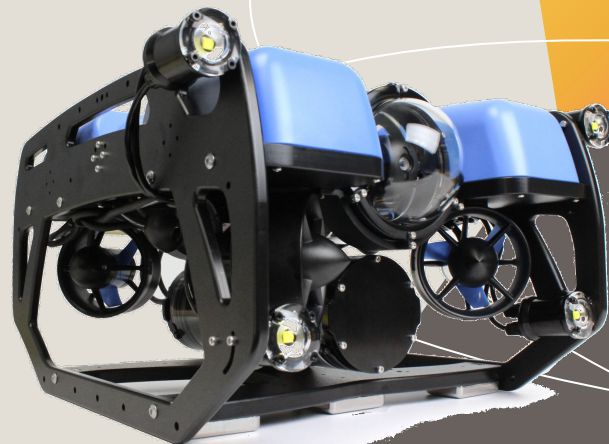
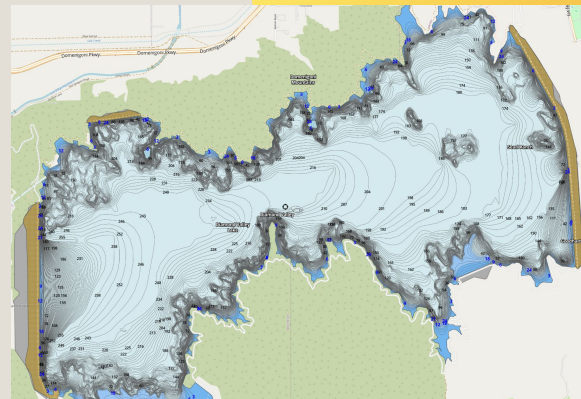
Bluefin

Why Bluefin?

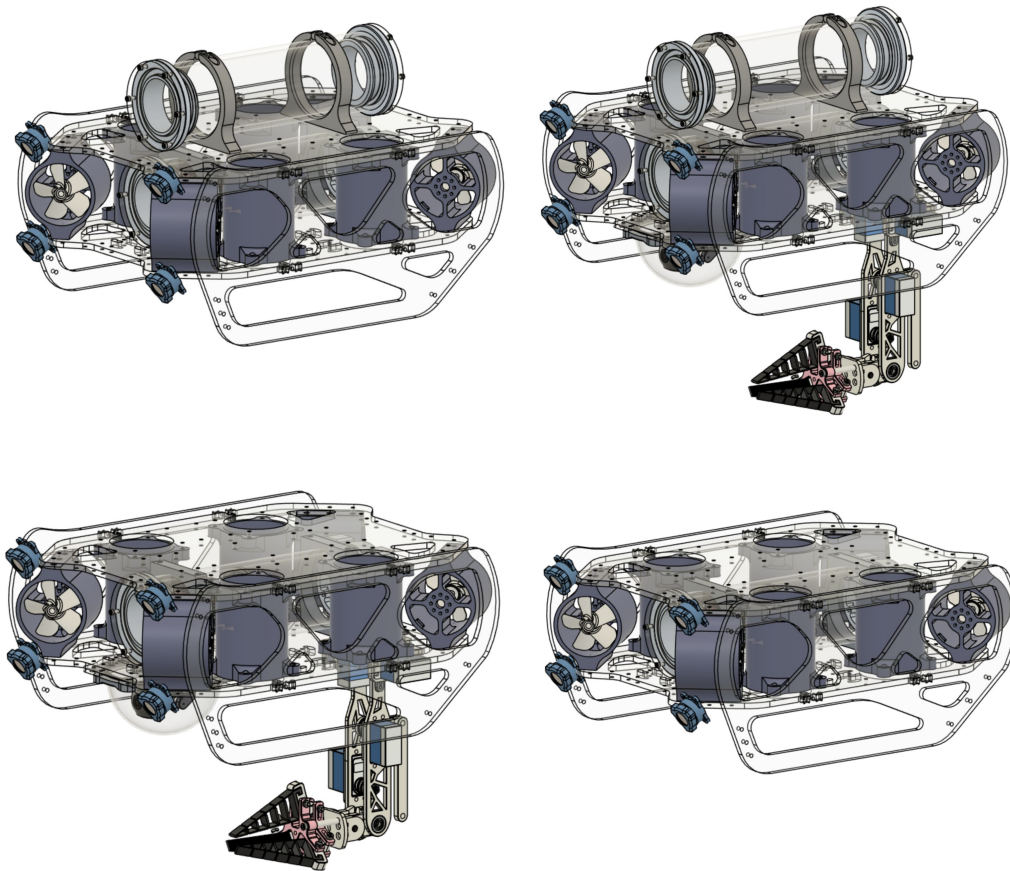


Our Use Case at Webb

- This project began after the Alf Museum wanted a way to explore a few lake beds for fossils
- With many conditions being unknown due to low visibility an ROV (remote operated vehicle) was the best option to safely search for these.
- They ended up using the BlueROV, a popular option that has 2 configurations



Purpose: open-source, low-cost ROV platform



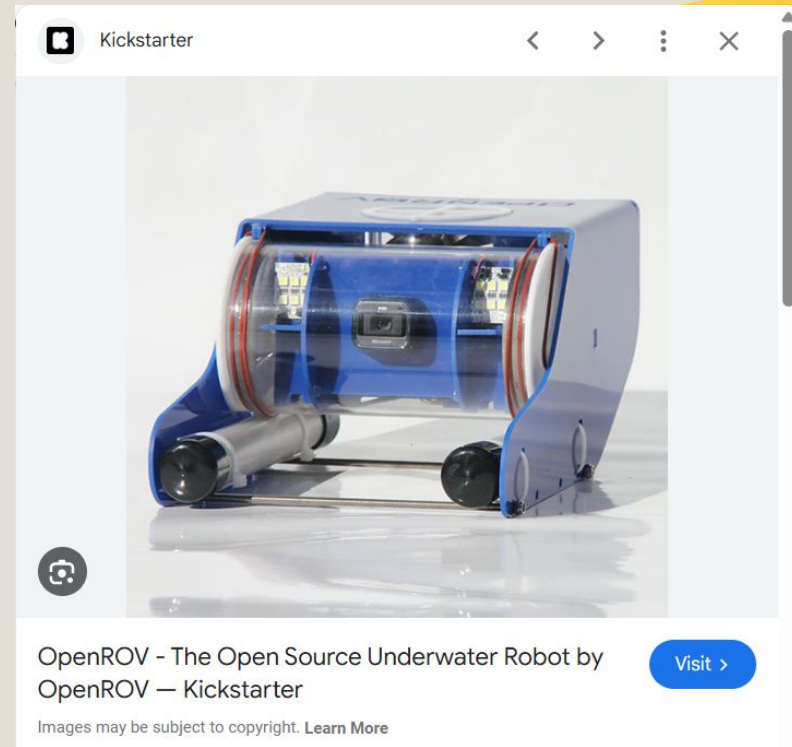
What do we mean by platform?

- A base chassis with many configurations using just the base parts
 - Can be used for long range explorations, or even just pool cleaning
- Modular mounting points mean more add-ons can be designed for many different use-cases

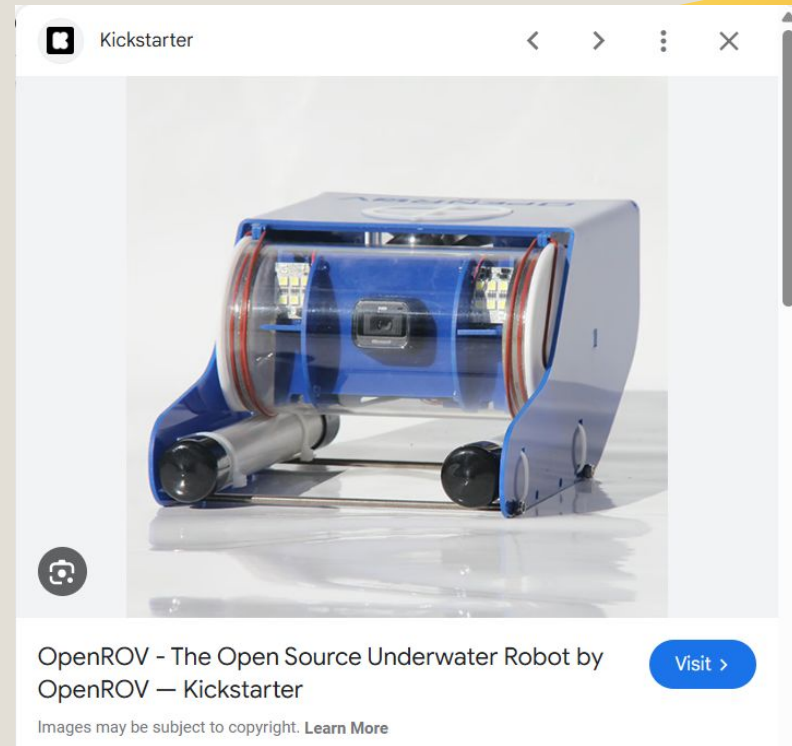
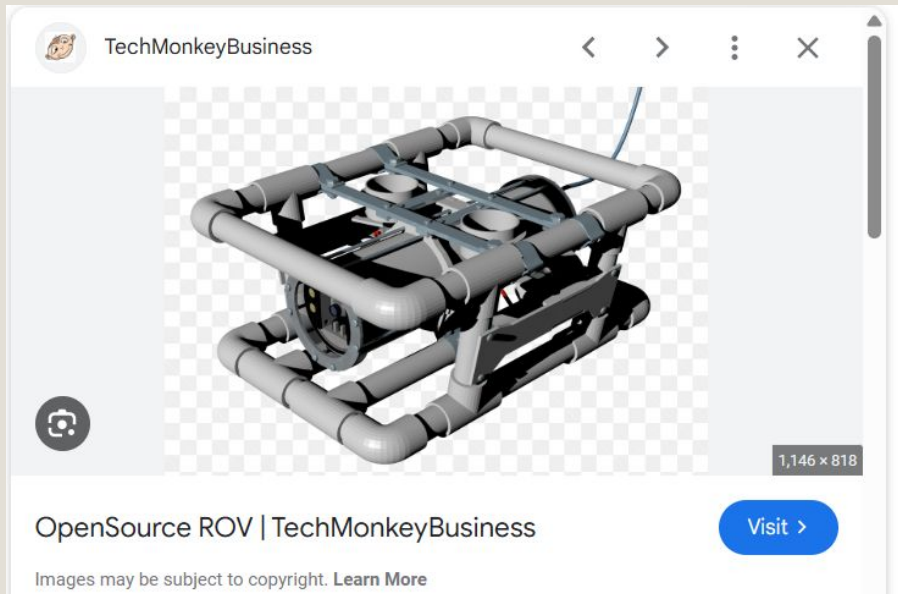
Cost Breakdown:

Polycarbonate frame	- \$240
Thruster motors	- \$176
Pixhawk	- \$317
Raspberry Pi	- \$35
4in1 ESC	- \$44
LED Lights	- \$23
Acrylic Tubes	- \$28
18650 Batteries	- \$36
3d-printed parts	- ~\$40
Rough Total	: \$939 (under a quarter of the cost of BlueROV)

Other options:



Other options:



Other options:



TechMonkeyBusiness





OpenSource ROV | TechMonkeyBusiness

Images may be subject to copyright. [Learn More](#)

Visit >



Hackaday





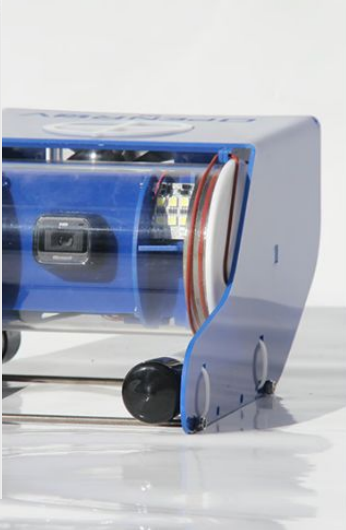
Aruna: An Open Source ROV For Affordable Research | Hackaday

Images may be subject to copyright. [Learn More](#)

Visit >



Kickstarter

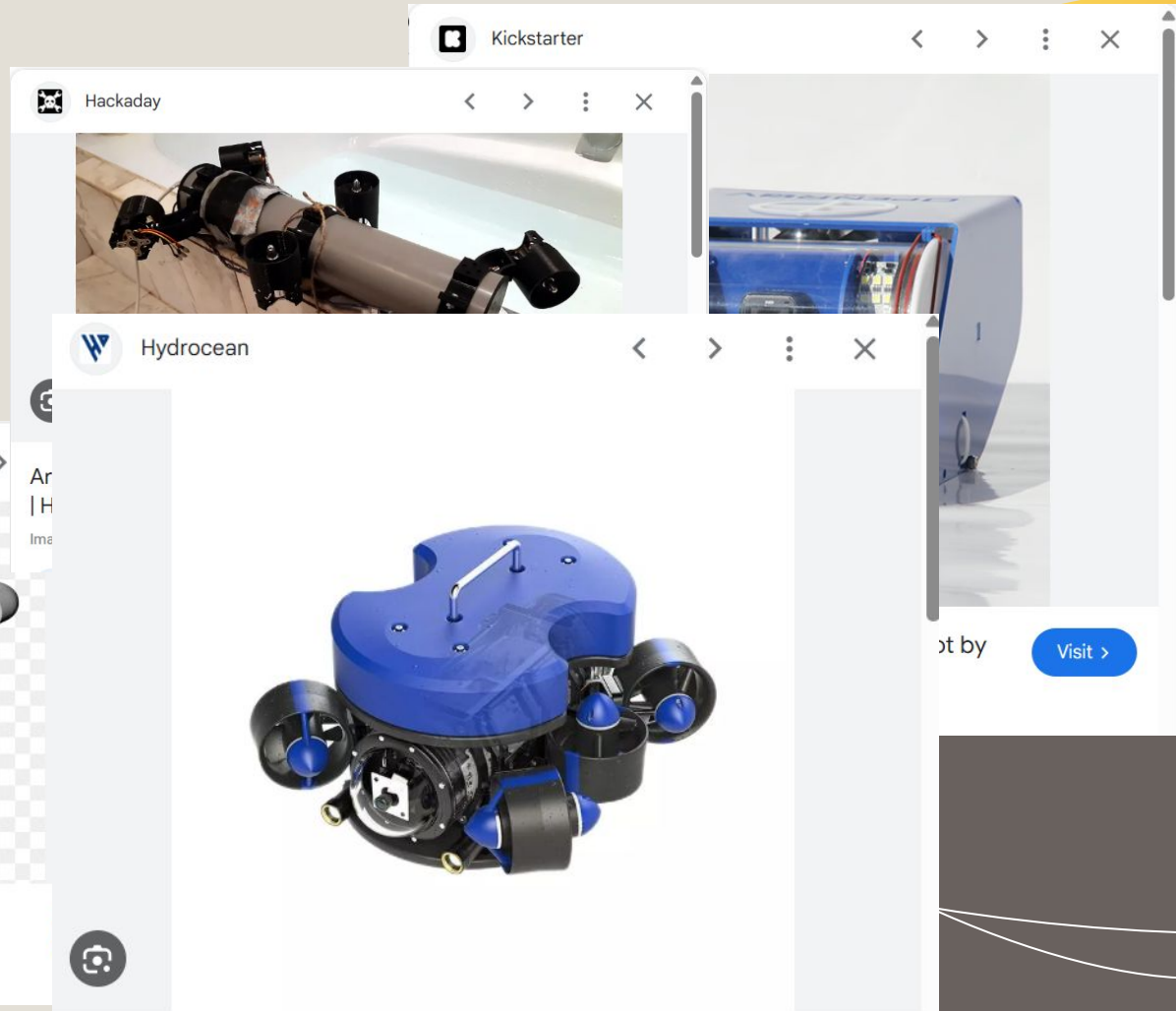
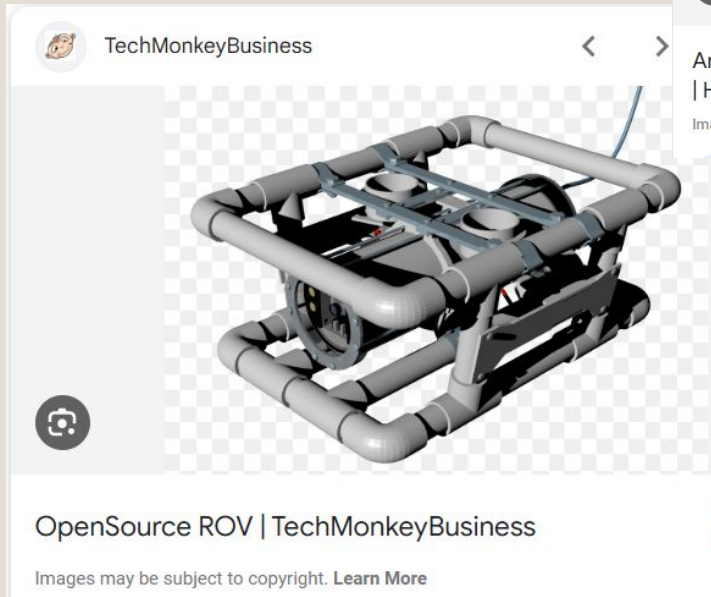


OpenROV - The Open Source Underwater Robot by OpenROV — Kickstarter

Images may be subject to copyright. [Learn More](#)

Visit >

Other options:



Other optio



TechMonkeyBusiness



OpenSource ROV | TechMonkeyBusiness

Images may be subject to copyright. [Learn More](#)



Influential Drones



SeaOtter 2 Underwater Drone / Submersible ROV

\$20,995.00* · [In stock](#) · Brand: JW Fishers

[Visit >](#)



Kickstarter

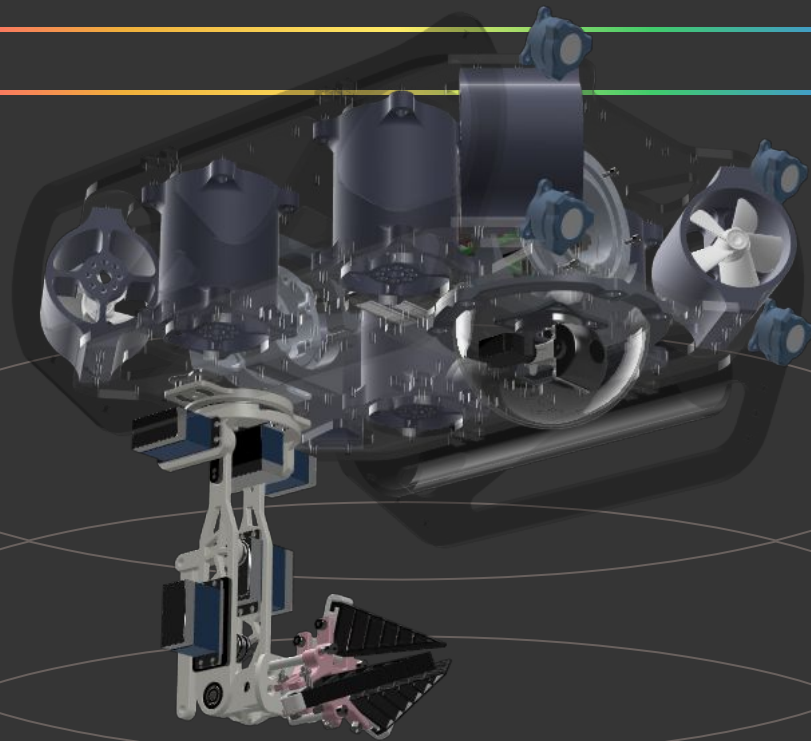
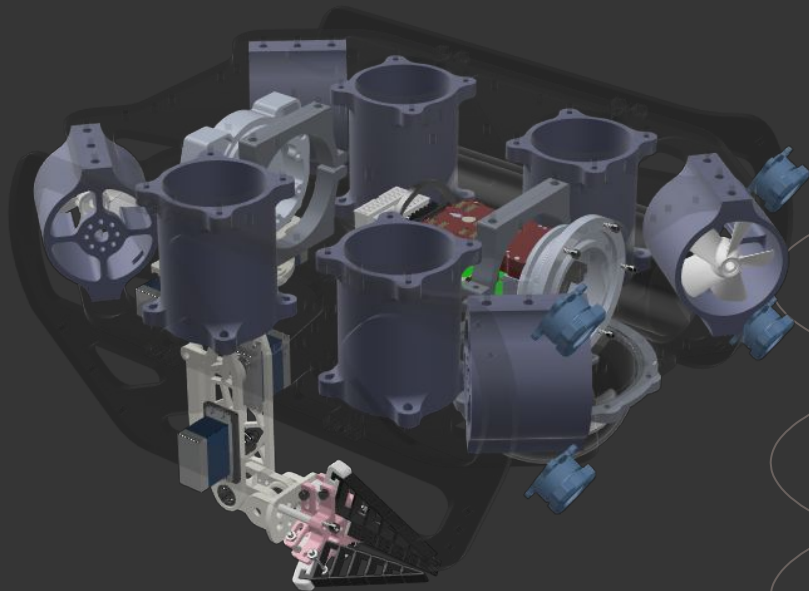


ot by

[Visit >](#)

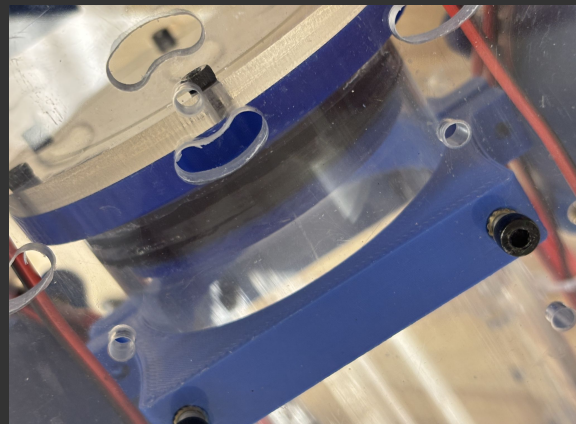
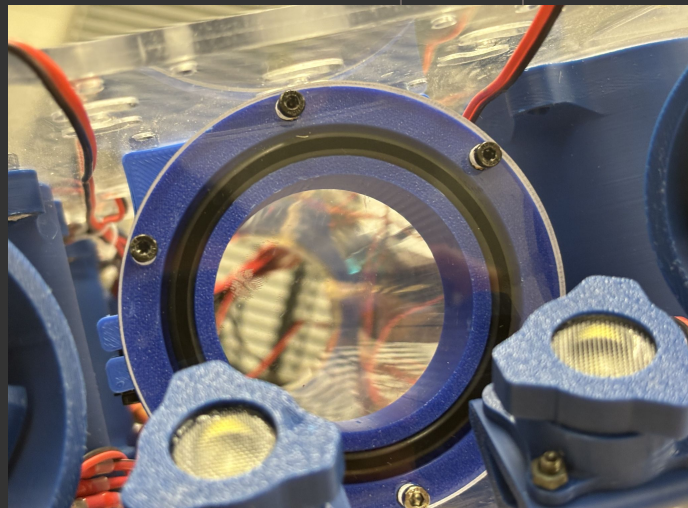
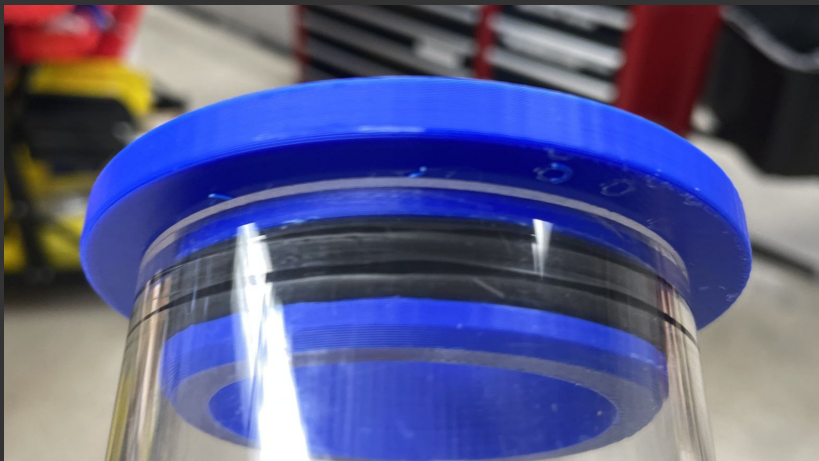
Bluefin

Build Progress

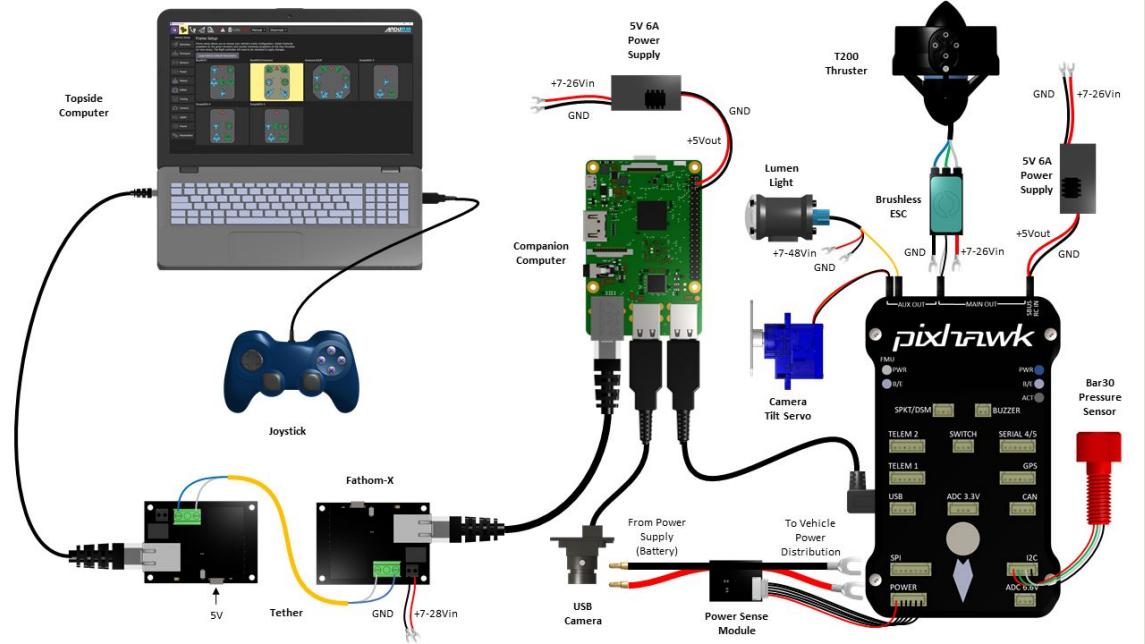
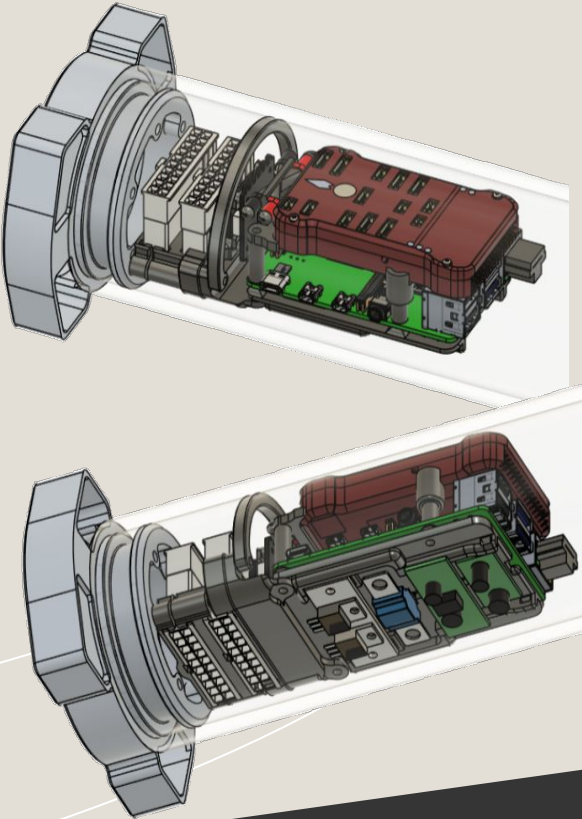


Bluefin

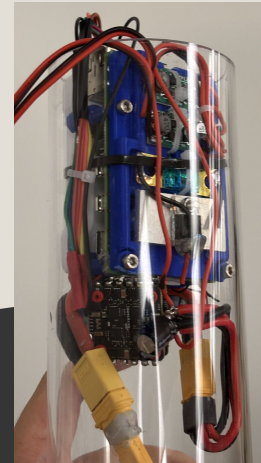
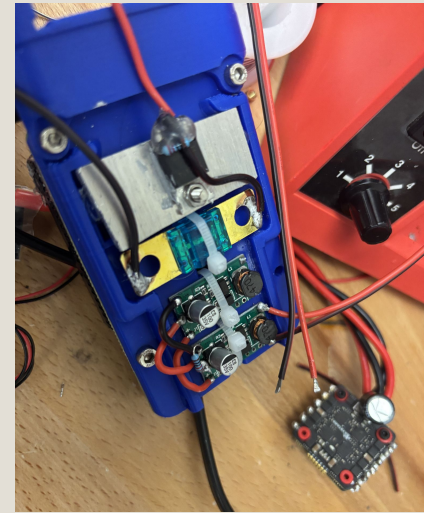
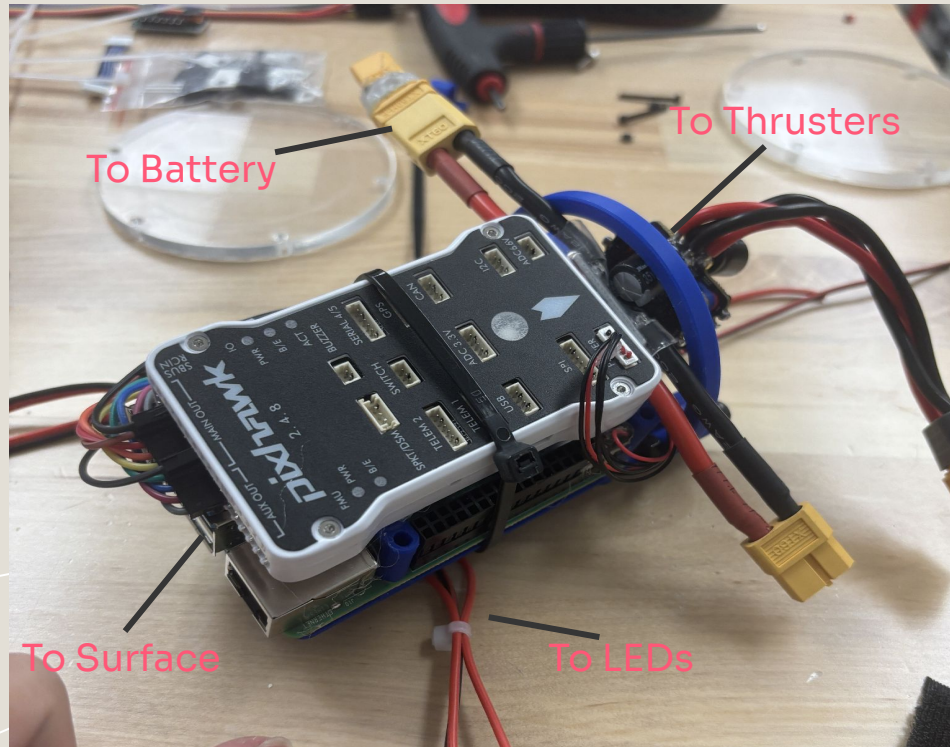
End Caps



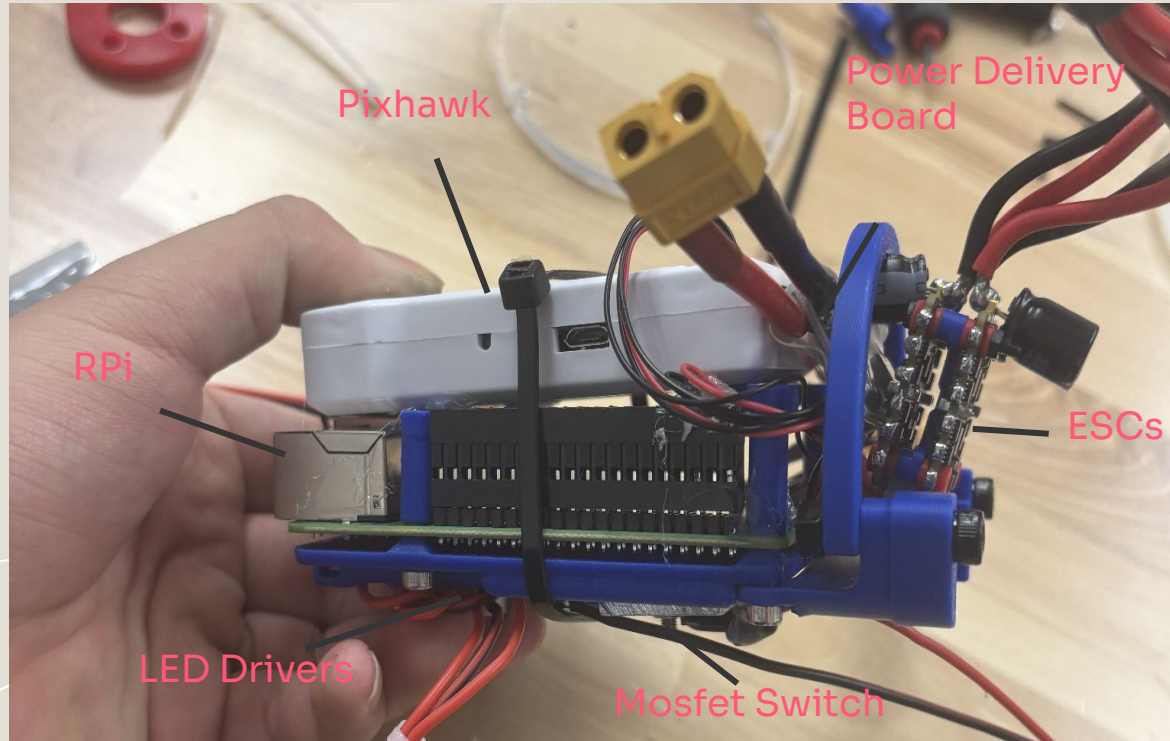
Electronics



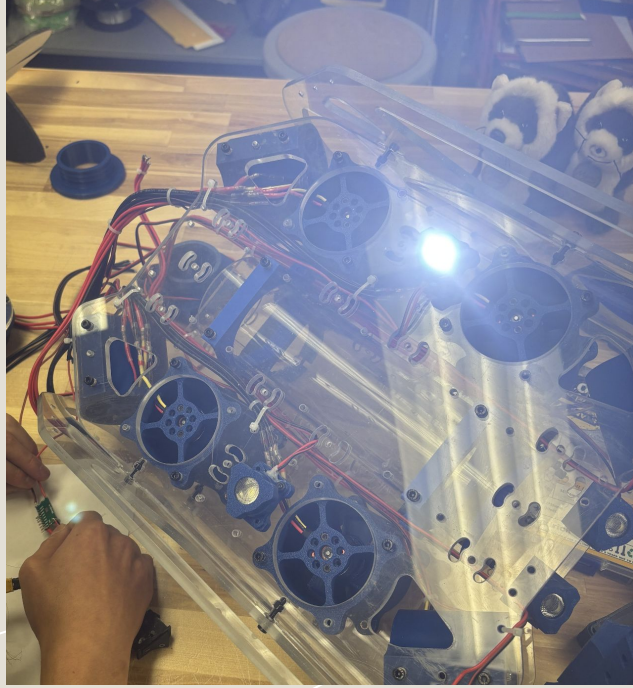
Electronics - continued



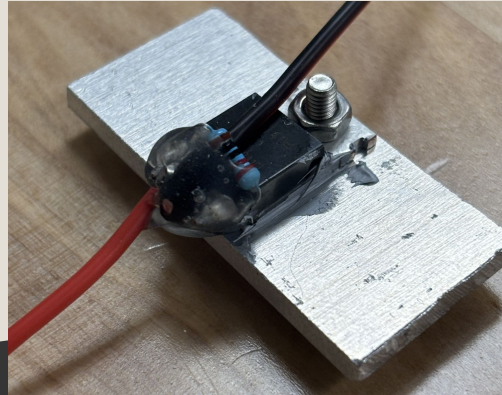
Electronics - continued



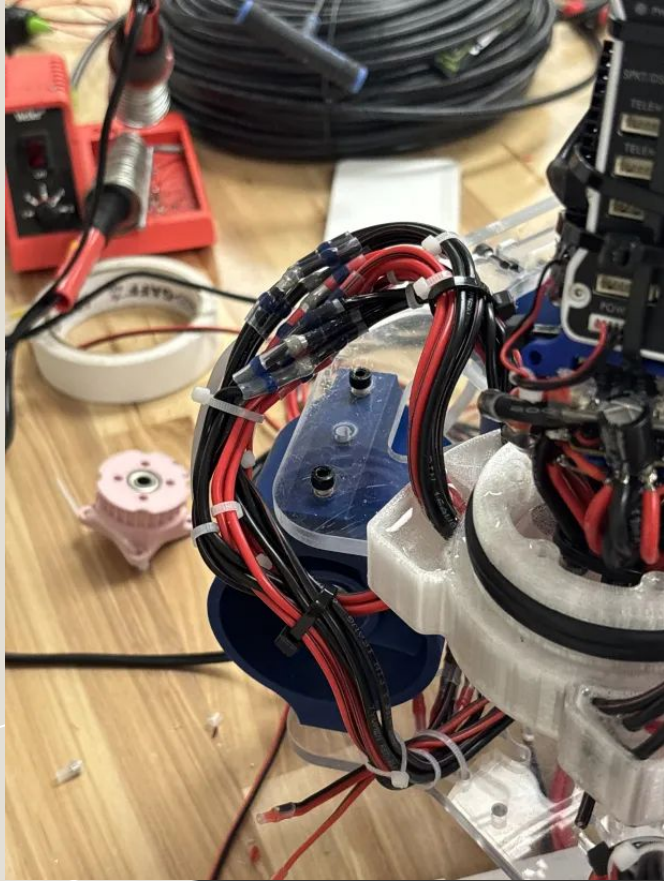
Batteries and Lighting



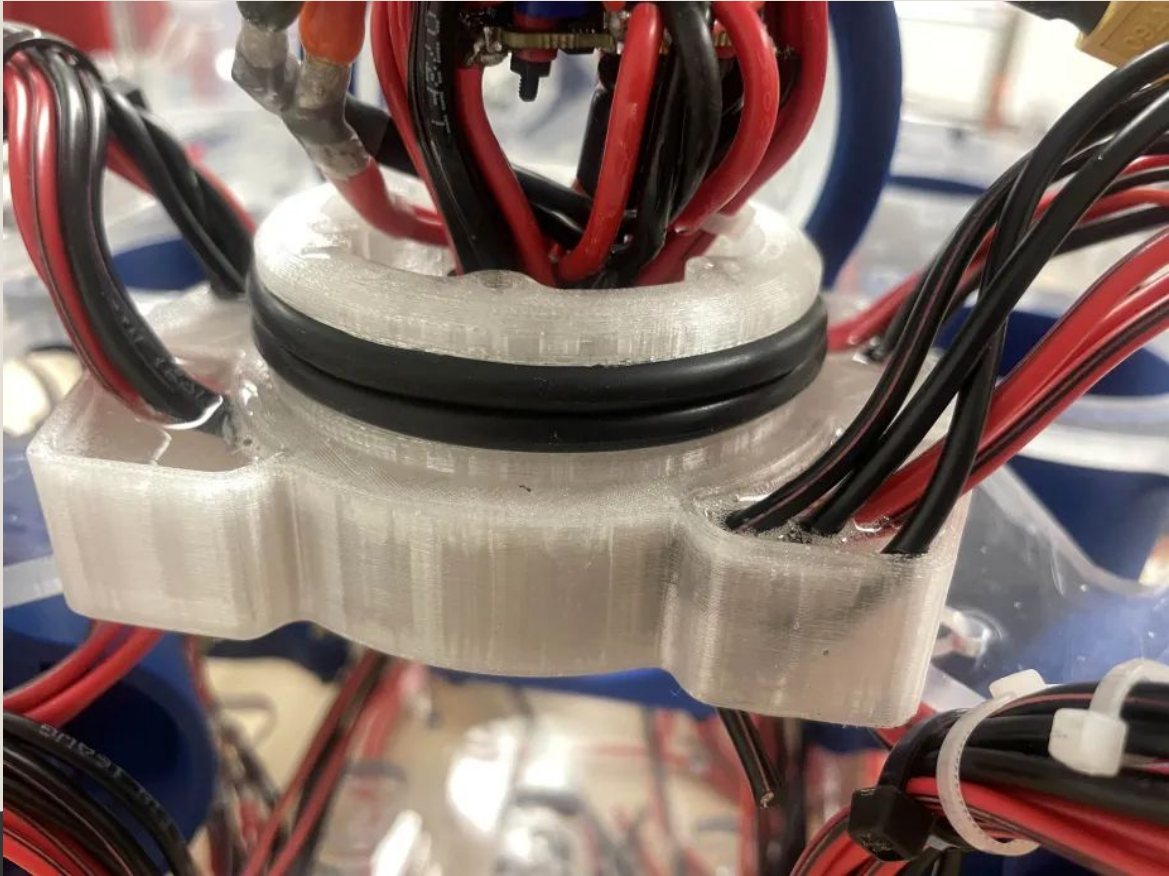
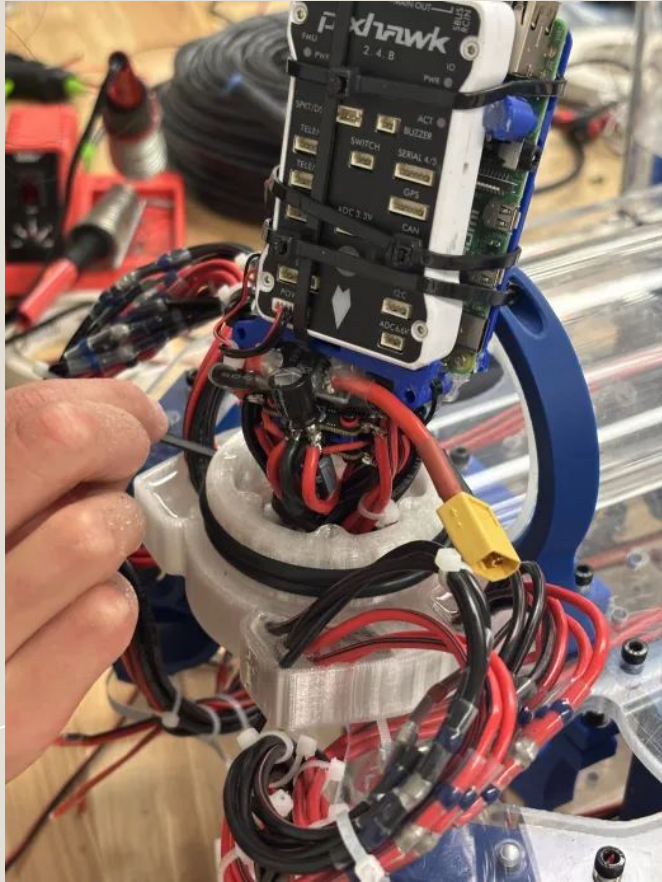
- Decision to bypass the mosfet switch for ease of wiring
- Testing capabilities of LED Drivers



Waterproofing Cables



Sealing Endcaps



Next Steps

- ~~Finish wiring to thrusters and LEDs~~
- An in-water test with all the subsystems installed
- Test effectiveness of new seals
- Test propeller and thruster effectiveness

Initial Waterproof test





Thank you